

Problem 1.)

- a.) d = impeller ϕ at angular velocity Ω
 ρ = liquid density
 P_1 = pressure in reservoir
 P_2 = pressure at pump exit
 Q = Volumetric flow rate

let ΔP = pressure rise = $f(\rho, Q, \Omega, d)$.

	ΔP	ρ	Q	Ω	d
M	1	1	0	0	0
L	-1	-3	3	0	1
T	-2	0	-1	-1	0

$\det = 0 \Rightarrow \text{Rank} = 3$

$Q = N - R = 5 - 3 = 2 \Rightarrow 2$ dimensionless groups

$\pi_1 = \Delta P \rho^a \Omega^b d^c$

$$\begin{cases} M: 1 + a = 0 \\ L: -1 - 3a + c = 0 \\ T: -2 - b = 0 \end{cases} \Rightarrow \begin{cases} a = -1 \\ b = -2 \\ c = -2 \end{cases} \Rightarrow \pi_1 = \Delta P \rho^{-1} \Omega^{-2} d^{-2}$$

$\pi_2 = Q \rho^a \Omega^b d^c$

$$\begin{cases} M: a = 0 \\ L: 3 - 3a + c = 0 \\ T: -1 - b = 0 \end{cases} \Rightarrow \begin{cases} a = 0 \\ b = -1 \\ c = -3 \end{cases} \Rightarrow \pi_2 = Q \Omega^{-1} d^{-3}$$

$$\Rightarrow F(\pi_1, \pi_2) = 0 \Rightarrow \frac{\Delta P}{\rho \Omega^2 d^2} = f\left(\frac{Q}{\Omega d^3}\right)$$

Where: $H \triangleq \frac{\Delta P}{\rho g}$

$$\Rightarrow \frac{gH}{\Omega^2 d^2} = f\left(\frac{Q}{\Omega d^3}\right)$$

Problem 2.)

a.) $\varepsilon = f(p, T, m, k_B, h, c)$

	ε	p	T	m	k_B	h	c
M	1	1	0	1	1	1	0
L	-1	-3	0	0	2	2	1
T	-2	0	0	0	-2	-1	-1
T	0	0	1	0	-1	0	0

$Q = N - R = 7 - 4 = 3$ dimensionless groups

$\pi_1 = \varepsilon m^a k_B^b h^c c^d$

$M: 1 + a + b + c = 0$
 $L: -1 + 2b + 2c + d = 0$
 $T: -2 - 2b - c - d = 0$
 $T: -b = 0$

$\Rightarrow \begin{cases} a = -4 \\ b = 0 \\ c = 3 \\ d = -5 \end{cases}$

$\Rightarrow \pi_1 = \varepsilon m^{-4} h^3 c^{-5}$

$\pi_2 = p m^a k_B^b h^c c^d$

$M: 1 + a + b + c = 0$
 $L: -3 + 2b + 2c + d = 0$
 $T: -2b - c - d = 0$
 $T: -b = 0$

$\Rightarrow \begin{cases} a = -4 \\ b = 0 \\ c = 3 \\ d = -3 \end{cases}$

$\Rightarrow \pi_2 = p m^{-4} h^3 c^{-3}$

$\pi_3 = T m^a k_B^b h^c c^d$

$M: a + b + c = 0$
 $L: 2b + 2c + d = 0$
 $T: -2b - c - d = 0$
 $T: 1 - b = 0$

$\Rightarrow \begin{cases} a = -1 \\ b = 1 \\ c = 0 \\ d = -2 \end{cases}$

$\Rightarrow \pi_3 = T m^{-1} k_B c^{-2}$

$\therefore \pi_1 = F(\pi_2, \pi_3) \Rightarrow \frac{\varepsilon h^3}{m^4 c^5} = f\left(\frac{p h^3}{m^4 c^3}, \frac{k_B T}{m c^2}\right)$

$\Rightarrow \varepsilon = \frac{m^4 c^5}{h^3} f\left(\frac{p h^3}{m^4 c^3}, \frac{k_B T}{m c^2}\right)$

b.) We require that h and c drop out of the formula.
 → Assume power-law form:

Hence: $F(\pi_2, \pi_3) = \sigma (\pi_2)^\alpha (\pi_3)^\beta$ Where: $\sigma \in \mathbb{R}$

$$\Rightarrow \varepsilon = \sigma \frac{m^4 c^5}{h^3} \left(\frac{p h^3}{m^4 c^3} \right)^\alpha \left(\frac{k_B T}{m c^2} \right)^\beta$$

$$\Rightarrow \varepsilon = \sigma p^\alpha T^\beta k_B^\beta m^{(4-4\alpha-\beta)} h^{(3\alpha-3)} c^{(5-3\alpha-2\beta)}$$

$$\Rightarrow \begin{cases} h: 3\alpha-3=0 \\ c: 5-3\alpha-2\beta=0 \end{cases} \Rightarrow \begin{cases} \alpha=1 \\ \beta=1 \end{cases}$$

$$\Rightarrow \boxed{\varepsilon = \sigma \frac{p k_B T}{m}} \quad \text{which follows the ideal gas law} //$$

c.) We require that m and p drop out of the formula.

$$\Rightarrow \begin{cases} m: 4-4\alpha-\beta=0 \\ p: \alpha=0 \end{cases} \Rightarrow \begin{cases} \alpha=0 \\ \beta=4 \end{cases}$$

$$\Rightarrow \boxed{\varepsilon = \sigma \frac{k_B^4 T^4}{h^3 c^3}} \quad \text{This is the Stefan-Boltzmann law} //$$

Problem 3.)

$$\partial_t \omega + u \partial_x \omega + v \partial_y \omega + w \partial_z \omega = -\frac{1}{\rho} \partial_z P - g + \nu (\partial_{xx} \omega + \partial_{yy} \omega + \partial_{zz} \omega)$$

a.) $x^* = \frac{x}{\ell}, y^* = \frac{y}{\ell}, z^* = \frac{z}{\ell}, u^* = \frac{u}{U}, v^* = \frac{v}{U}, \omega^* = \frac{\omega}{U},$
 $t^* = t \frac{U}{\ell}, p^* = \frac{p}{\rho U^2}, \partial_{\varphi^*} = \frac{\partial}{\partial \varphi \ell}, \partial_{\varphi^* \varphi^*} = \frac{\partial^2}{\partial(\varphi \ell) \partial(\varphi \ell)}$

$$\partial_t \omega = \ell^{-1} U^2 \partial_{t^*} \omega^*, u \partial_x \omega = \ell U u^* \partial_{x^*} \omega^*, v \partial_y \omega = \ell U v^* \partial_{y^*} \omega^*, w \partial_z \omega = \ell U^2 \omega^* \partial_{z^*} \omega^*$$

$$\partial_z P = \ell^{-1} \rho U^2 \partial_{z^*} P^*, \partial_{xx} \omega = U \frac{\partial^2}{\partial(x^* \ell) \partial(x^* \ell)} \omega^*, \dots$$

→ Divide by: $\left(\frac{U^2}{l}\right)$

$$\Rightarrow \partial_t^* \omega^* + U^* \partial_{x^*} \omega^* + V^* \partial_{y^*} \omega^* + \omega^* \partial_z^* \omega^* = -\partial_{z^*} P^* - g l U^{-2} + Re^{-1} (\partial_{x^* x^*}^2 \omega + \partial_{y^* y^*}^2 \omega + \partial_{z^* z^*}^2 \omega)$$

b.) $\frac{g l}{U^2} = \frac{1}{Fr^2} \equiv \text{Inverse Froude Number}$

$\frac{U}{\nu l} = \frac{1}{Re} \equiv \text{Inverse Reynolds Number}$

c.) $\omega = f(g, \mu, U, l)$

$$\begin{array}{c} \omega \quad g \quad \mu \quad U \quad l \\ M \quad 0 \quad 1 \quad 0 \quad 1 \quad 0 \quad 0 \\ L \quad 1 \quad -3 \quad 1 \quad -1 \quad 1 \quad 1 \\ T \quad -1 \quad 0 \quad -2 \quad -1 \quad -1 \quad 0 \end{array}$$

$Q = N - R = 6 - 3 = 3$ dimensionless groups

$$\pi_1 = \omega g^a U^b l^c$$

$$\begin{array}{l} M: a = 0 \\ L: 1 - 3a + b + c = 0 \\ T: -1 - b = 0 \end{array} \Rightarrow \begin{cases} a = 0 \\ b = -1 \\ c = 0 \end{cases} \Rightarrow \boxed{\pi_1 = \omega U^{-1}}$$

$$\pi_2 = g g^a U^b l^c$$

$$\begin{array}{l} M: a = 0 \\ L: 1 - 3a + b + c = 0 \\ T: -2 - b = 0 \end{array} \Rightarrow \begin{cases} a = 0 \\ b = -2 \\ c = 1 \end{cases} \Rightarrow \boxed{\pi_2 = g U^{-2} l}$$

$$\pi_3 = \mu g^a U^b l^c$$

$$\begin{array}{l} M: 1 + a = 0 \\ L: -1 - 3a + b + c = 0 \\ T: -1 - b = 0 \end{array} \Rightarrow \begin{cases} a = -1 \\ b = -1 \\ c = -1 \end{cases} \Rightarrow \boxed{\pi_3 = \mu g^{-1} U^{-1} l^{-1}}$$

$$\pi_2 = \frac{1}{Fr^2} \quad \text{and} \quad \pi_3 = \frac{1}{Re} \quad \Rightarrow \quad \frac{\omega}{U} = f(Fr, Re) = f\left(\frac{U}{\sqrt{g l}}, \frac{g U l}{\mu}\right)$$

d.) model $\equiv a_m$, prototype $\equiv a_p$ for any a

We have that: $Fr_m = Fr_p$, $Re_m = Re_p$, and $l_m/l_p = \frac{1}{100}$

$$\frac{U_m}{\sqrt{g l_m}} = \frac{U_p}{\sqrt{g l_p}} \Rightarrow \frac{U_m}{U_p} = \left(\frac{l_m}{l_p} \right)^{1/2} = \frac{1}{10}$$

$$\frac{U_m l_m}{\nu_m} = \frac{U_p l_p}{\nu_p} \Rightarrow \frac{U_m}{U_p} = \left(\frac{U_m}{U_p} \right) \left(\frac{l_m}{l_p} \right) = \left(\frac{l_m}{l_p} \right)^{1/2} \left(\frac{l_m}{l_p} \right) = \frac{1}{1,000}$$

$\rightarrow$ Suppose $\nu_p \approx 1 \times 10^{-6} \left[\frac{m^2}{s} \right]$ (water @ $20^\circ C$) $\Rightarrow \nu_m \approx 1 \times 10^{-9} \left[\frac{m^2}{s} \right]$